

HISTORA — Project Overview

A non-diagnostic oral-systemic research agent. One shared inflammatory proxy (excess IL-6 → CRP) links periodontal disease to cardiovascular, metabolic, and Alzheimer's disease; Claude orchestrates a calibrated, deterministic mechanistic engine to turn a fragmented case into **falsifiable, uncertainty-quantified, mechanistically-explained hypotheses** — validated on public data *and* genetics. It is a **research instrument a scientist can trust**, built with Claude for Life Sciences and co-organized with **Gladstone Institutes**. Hypothesis-generation, never a diagnosis or a treatment claim.

This document is the guided tour: the **problem** and **objective**, the **technical solution**, the **autonomous agentic workflow** we built and used to develop it, the **skill-evolution** capability we only partially exploited, the **flagship case run live in Claude Science**, and the **future directions** we set aside to focus on the hackathon.

Front-facing companions: [PITCH.md](#) (the thesis) · [CASE-STUDY.md](#) (the flagship session) · [PAPER.md](#) (the technical report) · [EVOLUTION.md](#) (self-improving skills) · [CLAUDE-SCIENCE.md](#) (run it in a real lab).

1. The problem and the objective

The problem

Gum disease, heart disease, diabetes, and Alzheimer's are studied in **separate silos** — different specialties, different datasets, different models. Yet a growing body of evidence suggests they may share one **upstream driver: chronic inflammation**, and specifically the cytokine **interleukin-6 (IL-6)**, whose hepatic output **C-reactive protein (CRP)** is the clinical footprint. Periodontitis is a lifelong, modifiable source of that inflammation.

Three things make this hard, and each is a design constraint for HISTORA:

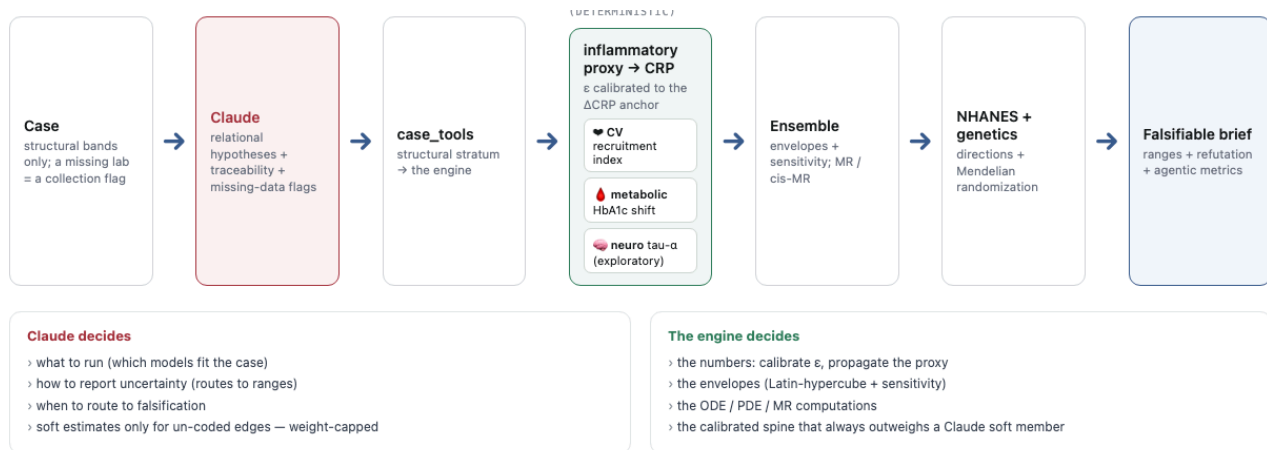
1. **Fragmentation.** Oral and systemic data live apart and are never co-present on one timeline, so the cross-domain relationships can't even be *represented*, let alone reasoned about.
2. **Small, honest effect sizes.** Periodontitis is *one contributor among many*. Any tool that inflates the signal to look impressive is scientifically dishonest and self-defeating.
3. **The AI honesty gap.** A language model asked to reason over health data will happily hallucinate an uncited number, give a point where only a range is warranted, or drift toward an individual diagnosis.

The objective

Build the **first safe, transparent, mechanistic research agent** for oral–systemic disease: an instrument that turns fragmented structural data into **one falsifiable research line**, where the honesty is not a disclaimer bolted on but a **structural property** — a *research-integrity gate* whose sharpest clause is **non-diagnostic** (population/parameter-level out, ranges not points, a missing datum is a collection flag never imputed, every claim traceable and falsifiable).

The target user is a **scientist working in Claude Science**, not a clinician and not a consumer. The win is not accuracy — it is **mechanism + honesty**: coherence (one lever, three diseases), calibration (to real

treatment data), and intellectual honesty (uncertainty, falsification, and the reported nulls as *features*).



Non-diagnostic throughout — Claude never sources a headline number; the deterministic engine is pinned, unit-tested code.

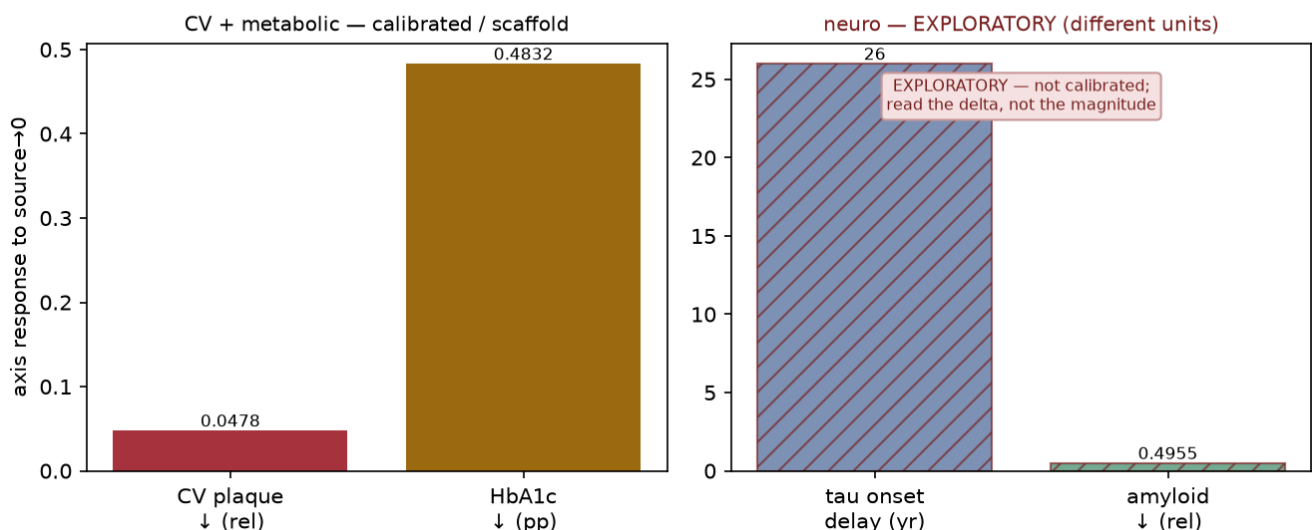
Who does what: Claude decides what to run, how to report uncertainty, and when to route to falsification; the deterministic engine decides the numbers. Claude never sources a headline number.

2. The technical solution

One shared proxy, three axes

The spine is a single calibrated parameter. A periodontal source raises **IL-6**; IL-6 → **CRP** is the one uncertain edge, pinned (ε) to a *real interventional anchor* — the meta-analytic **ΔhsCRP after periodontal therapy (~0.5 mg/L)**. From that shared inflammatory gain, three axes fork **coherently**: one intervention moves all three together, not three independent guesses.

One lever, many axes — periodontal therapy's coherent multi-axis response

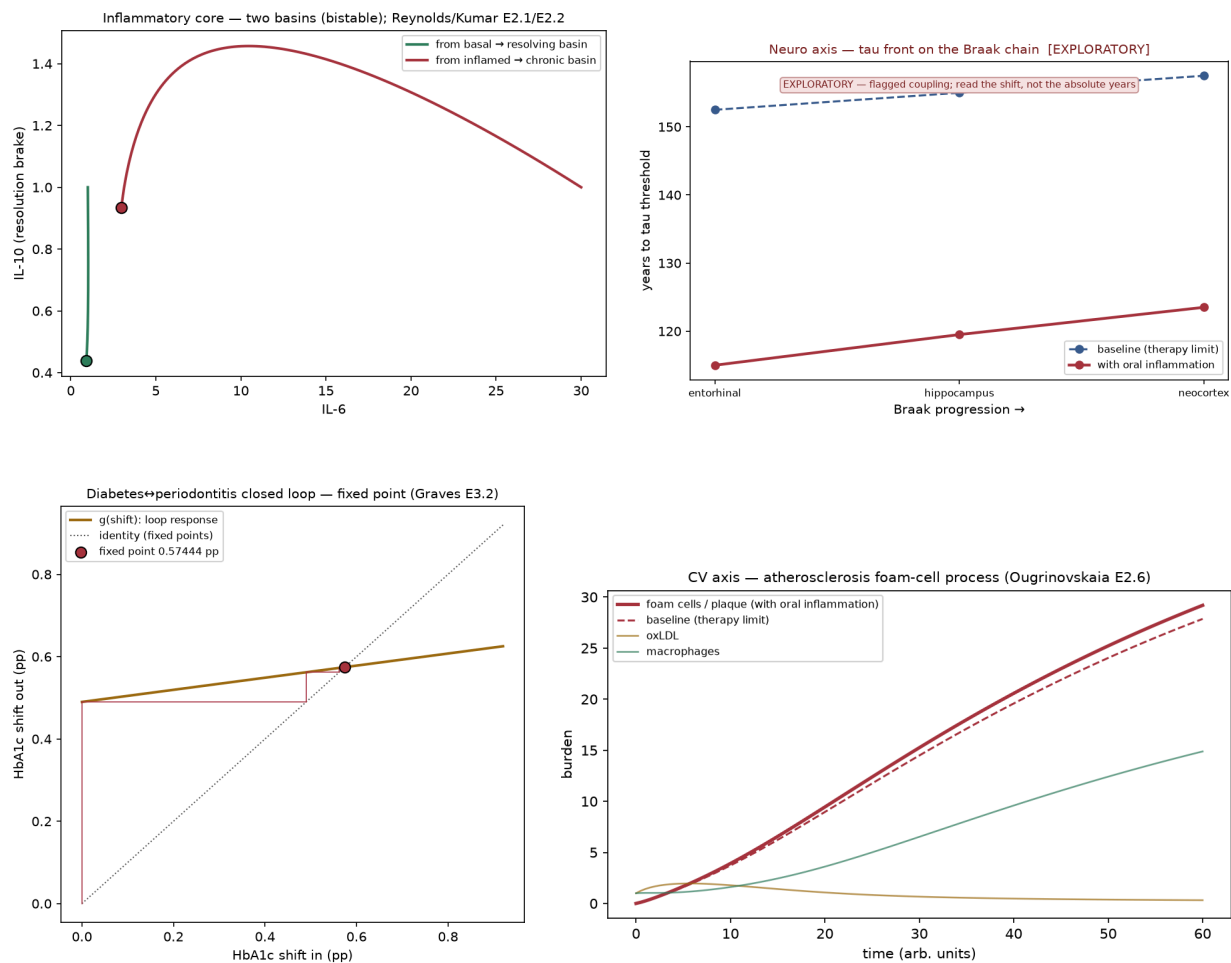


The integrative panel. The neuro axis is drawn separately and marked EXPLORATORY so its numbers can never be quoted as a calibrated result — the honesty is in the figure, not just the caption.

Mechanistic depth (Stage-3)

Each axis is a real, tested, non-diagnostic model — not a linear multiplier:

Axis	Model	What it shows
Inflammatory core	reduced Reynolds/Kumar TNF/IL-6/IL-10 ODE	two basins — acute/resolving vs chronic (bistability the single scalar can't express)
Cardiovascular	Ougrinovskaia foam-cell atherosclerosis ODE	plaque as a process , not an index
Metabolic	Bergman glucose–insulin minimal model	S _I degraded by inflammation → HbA1c, calibrated to the ~0.35 pp therapy anchor
Diabetes↔perio	closed feedback loop	a fixed point — hyperglycaemia amplifies the source, the source raises the gain
Neuro (<i>exploratory</i>)	Hao–Friedman amyloid + Braak tau front	the A/T cascade, flagged , APOE4/age as effect modifiers

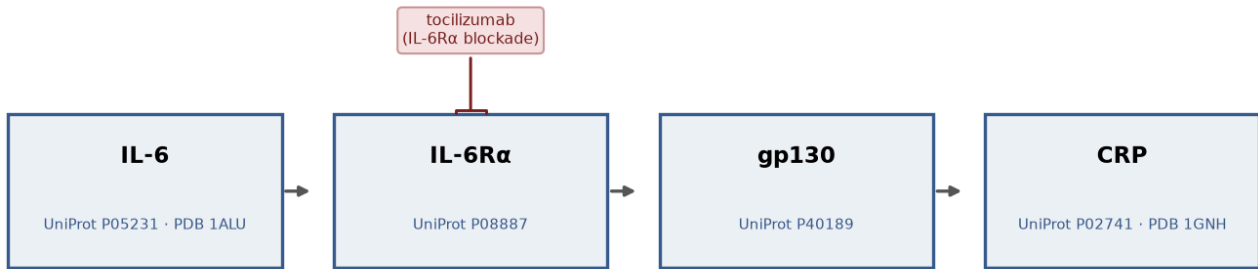


Left→right: the two inflammatory basins; the tau front advancing on the Braak chain (marked *EXPLORATORY* — read the shift, not the absolute years); the diabetes↔perio loop converging to a fixed point; the foam-cell plaque process.

The protein layer — grounding the mechanism in molecules

The shared proxy is grounded in real molecular identity: **IL-6** → **IL-6Rα** / **gp130** → **CRP**, with **tocilizumab** (an IL-6Rα-blocking antibody) marking the causal node. Every mediator carries its **UniProt accession** — the stable key the Claude Science UniProt/PDB connector resolves into live 3-D structures.

Protein layer — the shared-proxy signaling axis (reference IDs)



periodontal therapy lowers IL-6 at the same node tocilizumab blocks · live 3-D structures via the Claude Science UniProt/PDB connector

Validation — the "whether", independent of the calibration

- **NHANES** (public, de-identified): the three predicted directions are real and **survive design-adjusted (survey-weighted + FDR)** statistics.
- **Mendelian randomization** (public GWAS): **IL-6R → coronary disease is causal**; **circulating CRP → CAD is null** — the marker isn't causal, the *node* is; and **CRP/IL-6 → Alzheimer's is null**, which is why the neuro axis stays exploratory.

Calibration ≠ validation. Calibration pins the one uncertain edge to an *interventional* anchor; validation is the independent NHANES signs and the genetic probe. We never present one as the other.

The research-integrity gate (non-diagnostic by construction)

Every output is checked against a **protected invariant**: no diagnosis or individual claim; no imputed patient value (missing → collection flag); evidence-grounded and schema-valid; population/parameter-level. This gate is enforced as a hard binary check and — crucially — it sits **outside** the part of the system Claude is allowed to self-edit (see §4).

Delivery — two surfaces, one engine

A portable **Claude Code plugin** (the demo a judge runs anywhere) and its lab-grade home **Claude Science**, where the same `skills/` become skills, the `histora` harness a reusable pipeline, the `agents/` specialist agents, and UniProt/PDB/OpenGWAS/NHANES the connectors — the same components, ported directly.

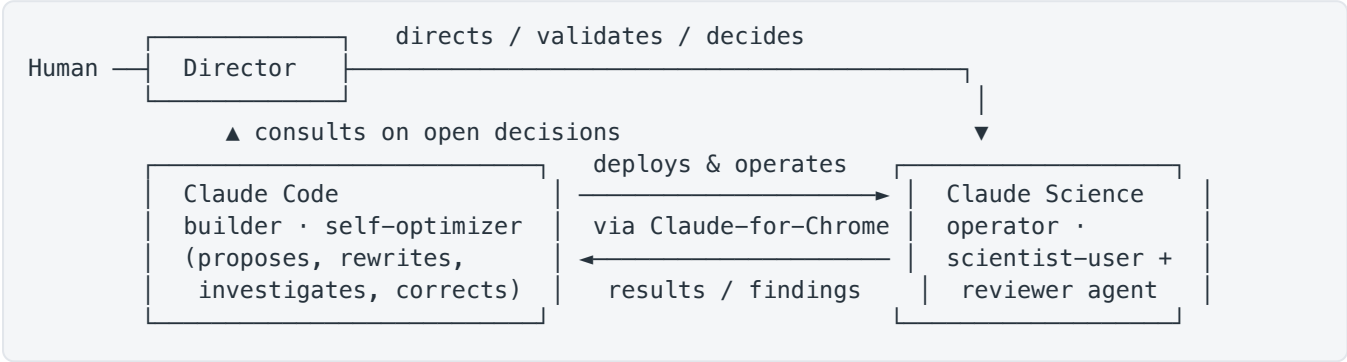
3. The development workflow — an autonomous agentic system

The build itself is an artifact of this project. HISTORA was developed and operated by an **autonomous agentic system** under human direction, exercising Claude at **every layer**:

- **Director (human).** A bioengineer + AI engineer sets the goals, validates, makes the judgment calls, and steers — approving risks, choosing anchors, and vetoing scope.
- **Builder + self-optimizer (Claude Code).** An autonomous agent that **proposes, rewrites its own skills, investigates, self-corrects, and consults** the director on genuinely open decisions. It authored

the engine, the tests, the docs, and the SkillOpt loop, iterating through pull requests.

- **Operator (Claude Code → Claude-for-Chrome → Claude Science).** The same agent **deploys and operates** the project: it drives the Claude-for-Chrome plugin to run HISTORA inside Claude Science (a local web app), importing the skills, reinstalling the engine, running the pipeline, and driving the UniProt/PDB and OpenGWAS connectors.
- **Scientist-user (Claude, inside Claude Science).** Claude acts as a **qualified user** that evaluates the work, renders the figures and 3-D structures, and — through the **reviewer agent** — audits the outputs, *finds real flaws*, and prompts corrections.



Every claim above is **evidence-linked, not narrated**: the SkillOpt archive (§4), the live Claude Science run (§5), and the reviewer agent's real audit findings. This meta-role — Claude as builder, self-optimizer, operator, *and* scientist-user — is the differentiator for a **builder track**; the science leads, and this workflow is what reframes it as *"and Claude built and operates all of this, too."*

4. Skill evolution (SkillOpt) — a capability we only partially exploited

HISTORA can **improve its own reasoning skills**, safely. **SkillOpt** is a gated evolutionary loop: Claude proposes an edit to one of its trainable `SKILL.md` files; the edit is **adopted only** if it *measurably* improves a machine-checkable structural metric (bootstrap CI excludes 0) **AND** the non-diagnostic guardrail still passes on **every** case (a hard binary gate). Every generation — adopted and rejected — is appended to an **auditable archive** with parent→child lineage and the **guardrail hash**, identical in parent and child: machine-checkable proof the safety invariant never moved.

Why it's safe by construction: the *genome* is only the prose of the trainable skills; the guardrail, the citation registry, and the engine are **outside** it — evolution literally cannot reach them.

What we ran (the honest portfolio): live, with Claude as the mutation operator and a Claude agent as the scored reasoner, across three real skills —

Skill	Metric	Outcome	Pattern
traceability-audit	field-citation coverage	adopted 0.00 → 0.93	knowledge_gap
cardiometabolic-framing	pathway-tag coverage	adopted 0.00 → 0.67	execution_gap
record-normalization	MISSING-flag recall	null — parent already optimal	—

Two skills improved *by different mechanisms*; one was **correctly left alone** because it was already at ceiling. That the loop **doesn't manufacture a gain where none exists** is the anti-reward-hacking property that makes the adopted gains credible.

What we deliberately left out (hackathon time + safety discipline): we did **not** run multi-generation "optimize each skill to its maximum," and we did **not** build an ensemble co-optimization across all skills. Both are trainable in principle — but "optimize to the maximum" is, by definition, the reward-hacking behavior the whole design is built on *not* doing, and an ensemble-level metric would have to be model-judged or hand-designed, breaking the "fitness is external/structural" safety layer. We froze the portfolio at three: two adopted, one null. Extending this *safely* is a future direction (§6).

5. The documented case — run live in Claude Science

We ran the **flagship case** end-to-end inside Claude Science, on the current `main` engine, with the real connectors. This is *"scripted like a demo, real like an experiment"* — the path is real; nothing is faked.

The case and the research line (real engine output)

A structural stratum — **Stage III periodontitis · high BOP · type-2 diabetes · hs-CRP MISSING → collection flag**. The engine forks the shared proxy to three axes as **90% ranges**:

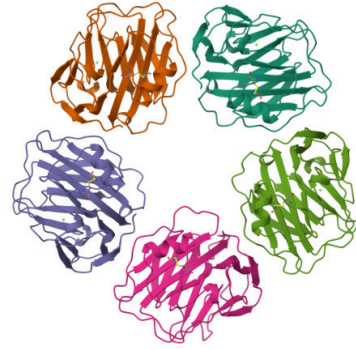
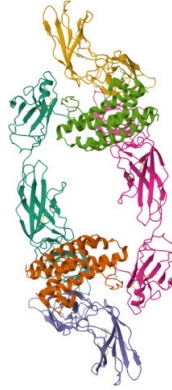
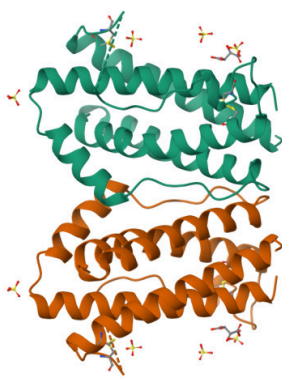
Output	Median	90% band
CRP (mg/L)	3.18	[2.89, 3.45]
HbA1c shift (pp)	+0.49	[0.27, 0.71]
tau-α rel. increase	+0.21	[0.11, 0.40] · EXPLORATORY

The falsifiable research line it generates (the safe anchor): - **Lever = periodontal therapy** (non-pharmacological, the exact edge the engine is calibrated on) — predicted **ΔhsCRP 0.68 mg/L**. - **Causal node = IL-6R**. cis-MR: **IL-6R → CAD causal** (LD-aware $\beta \approx +0.705$); **circulating CRP → CAD null**. **Tocilizumab** (IL-6Rα blockade) is cited **only** as proof the node is druggable/causal — **never as a treatment**. - **Hypothesis**: if the IL-6R node mediates the systemic effect, lowering the periodontal IL-6 source should move the markers in the direction the genetics implies — *a testable research line, not a therapy*. - **Ships its own refutation** (therapy fails to lower hs-CRP → calibration wrong; no IL-6R coronary effect → node premise fails; marker moves opposite → mediation refuted) and **names its shakiest assumption** (population→stratum transfer, untested).

Claude Science **verified every honesty red-line programmatically** against `case_study.json` ✓.

The molecular node — real 3-D structures via the UniProt/PDB connector

The connector resolved all six UniProt accessions and rendered real coordinates in the interactive Mol* viewer — including the **IL-6/IL-6Rα/gp130 hexameric signaling complex (PDB 1P9M, 5,322 atoms)**: the causal node itself, in one structure, where the IL-6R cis-MR instrument acts and tocilizumab blocks.



IL-6 ([1ALU](#) , four-helix bundle) → **IL-6/IL-6Ra/gp130 hexamer ([1P9M](#))** → CRP ([1GNH](#) , pentamer) — the causal node in real coordinates. These render **interactively in Claude Science** (via the UniProt/PDB connector) as the live 3-D upgrade of the static schematic in §2; the structure images above are the same PDB entries. *Structure images: RCSB PDB (rcsb.org).*

The scientist-user beat — the reviewer found a real (minor) error, and Claude self-corrected

The Claude Science **reviewer agent** audited the run and returned **one finding**: the closing prose stated the hexamer's resolution as "2.4 Å", but 1P9M is **3.65 Å** (2.4 Å belongs to [1N26](#) , a different, non-hexameric IL-6Ra structure). The saved manifest already recorded the correct 3.65 Å — only the chat prose slipped.

Claude read the finding and corrected itself immediately. Nothing downstream was affected.

This is the honesty loop working in the open: a qualified reviewer catches a real slip, the agent fixes it, and the record stays accurate — *shown, not dramatized*. (An earlier Stage-3 run had a similar honest finding, since resolved.)

6. Future directions — what we set aside to focus on the hackathon

Everything below is deliberately **out of scope for the hackathon** but is a natural, honest extension:

- **Safe multi-generation SkillOpt + ensemble.** Extend the loop beyond one gated generation *without* losing the anti-reward-hacking discipline — e.g. loop-until-dry with per-skill external metrics and an ensemble metric that stays structural/held-out, never model-judged.
- **More axes and mechanisms.** The model library holds cited-but-unbuilt models (renal, hepatic, further microbiome/keystone dynamics) ready to promote to tested code under the same tier discipline.
- **Live connectors at lab scale.** Run the harness and the UniProt/PDB/OpenGWAS/NHANES connectors at scale in Claude Science (a natural fit for Anthropic's AI-for-Science program), with native animated figures.
- **Secondary research lines (honest, non-leading).** Behavioral / prophylactic interventions that alter oral physiology; the *P. gingivalis* / gingipain → tau microbiome line (kept out of the lead because it adds new science and lives on the exploratory neuro axis) — as additional falsifiable hypotheses, never efficacy claims.
- **Prospective validation design.** The tool's whole purpose is to *pose* a study; a natural next step is to specify the design (cohort, endpoints, refutation criteria) that would test the IL-6R-mediation hypothesis it generates — moving from MR-supported *direction* toward interventional evidence.

- **Behavioral trace diagnostics.** Analyzing the agent's own short- and long-term reasoning traces to find and correct systematic defects — an internal-quality research track we scoped but did not build.
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Non-diagnostic throughout · population / parameter / molecular / instrument level only · MR ≠ RCT · calibration ≠ validation · hypothesis-generation, never efficacy, prevention, or diagnosis. Not medical advice — a research instrument.